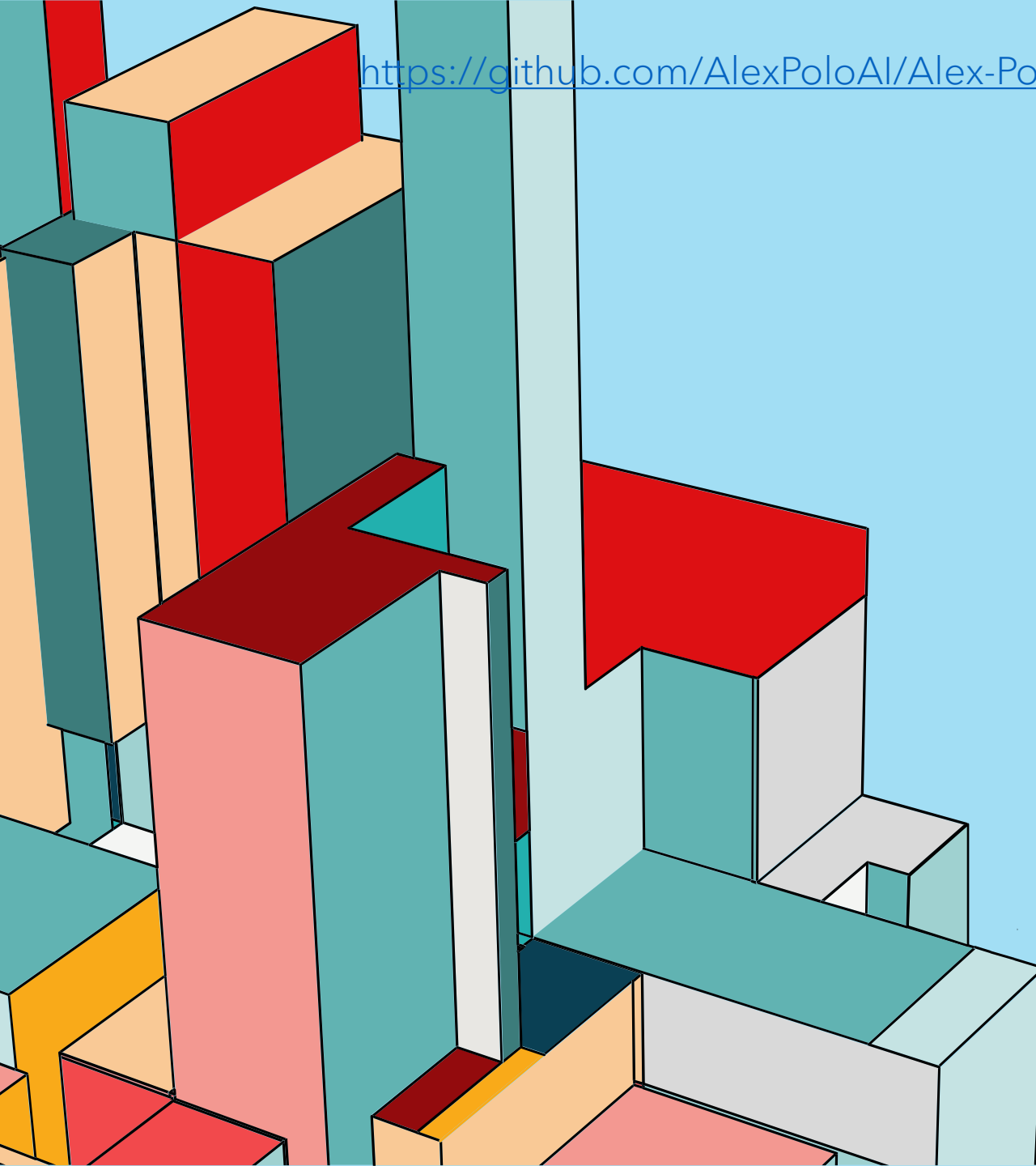


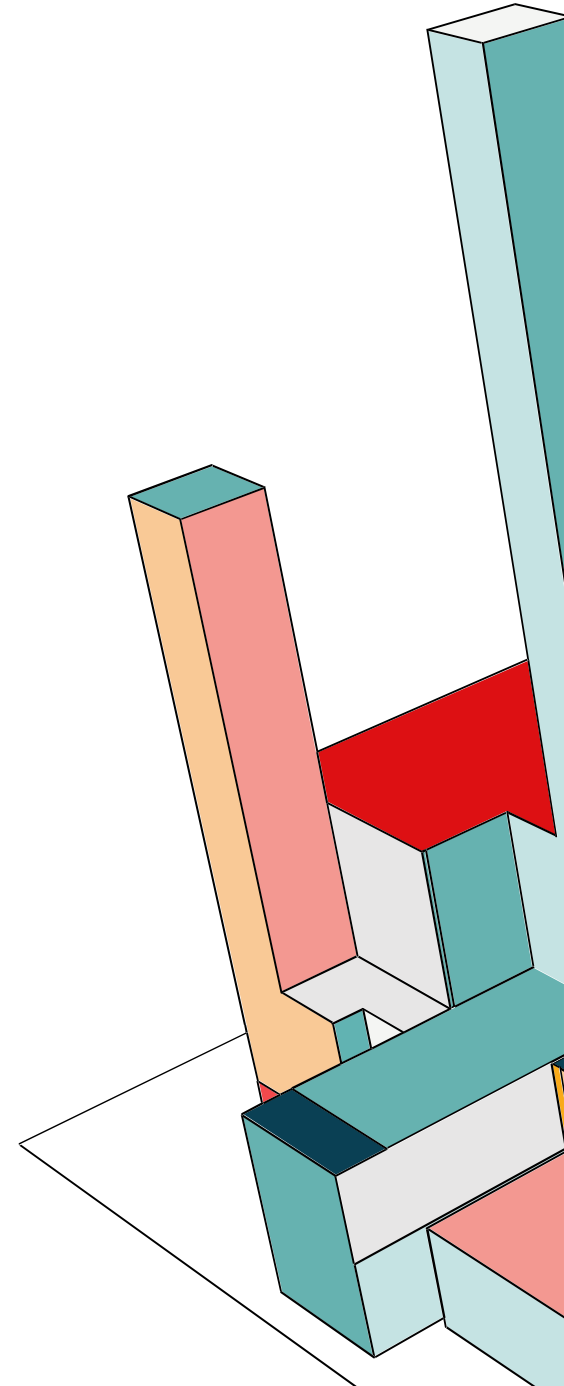


**PROJECT NAME:
ATTRACTOR PROJECT**

**DEVELOPER:
ALEX POLOZOV
TIER: 1**

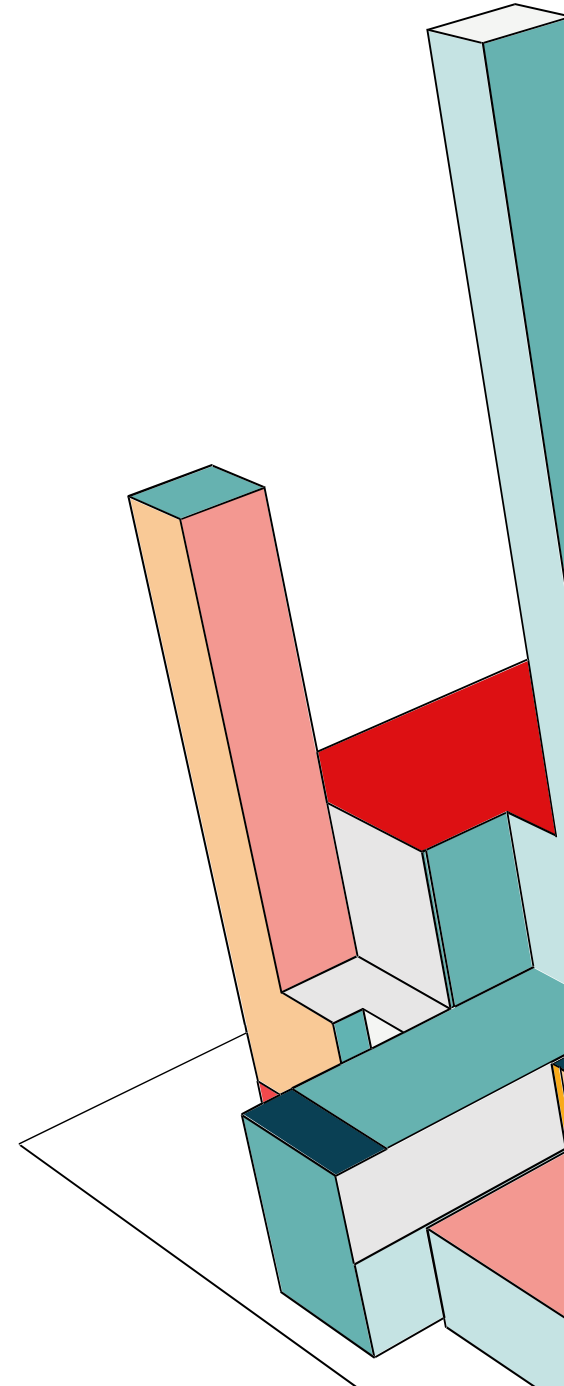
THE PROBLEM

- Generic algorithms cannot recognize subconscious human taste preferences. Standard biometric sorting fails when evaluating subjective aesthetic parameters in complex visual environments.



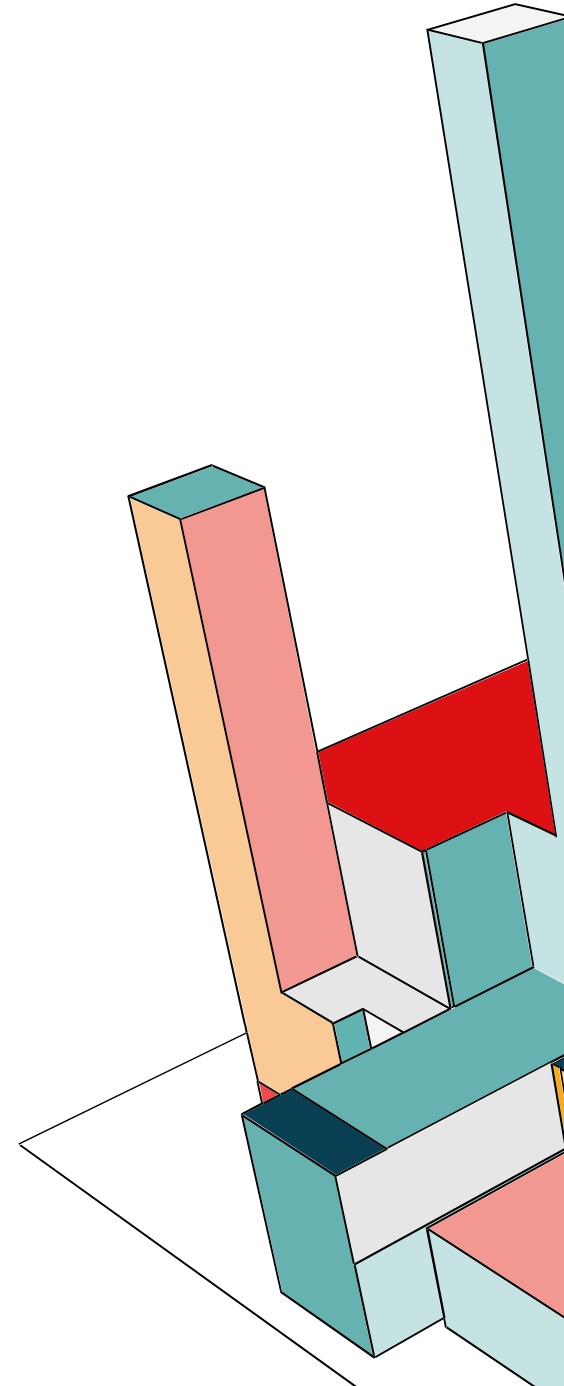
WHAT I BUILT

- I built a dual-pipeline computer vision system that filters images based on my personal visual taste matrix using both semantic and geometric approaches.
- **System Flow:**
 1. YOLO11x extracts humans from raw images to create local crops.
 2. **JSON Pipeline:** Qwen2-VL generates a semantic JSON matrix of physical and contextual attributes, scored by probability matching.
 3. **SVM Pipeline:** ResNet50 extracts feature vectors, and One-Class SVM measures geometric distances to rank images.



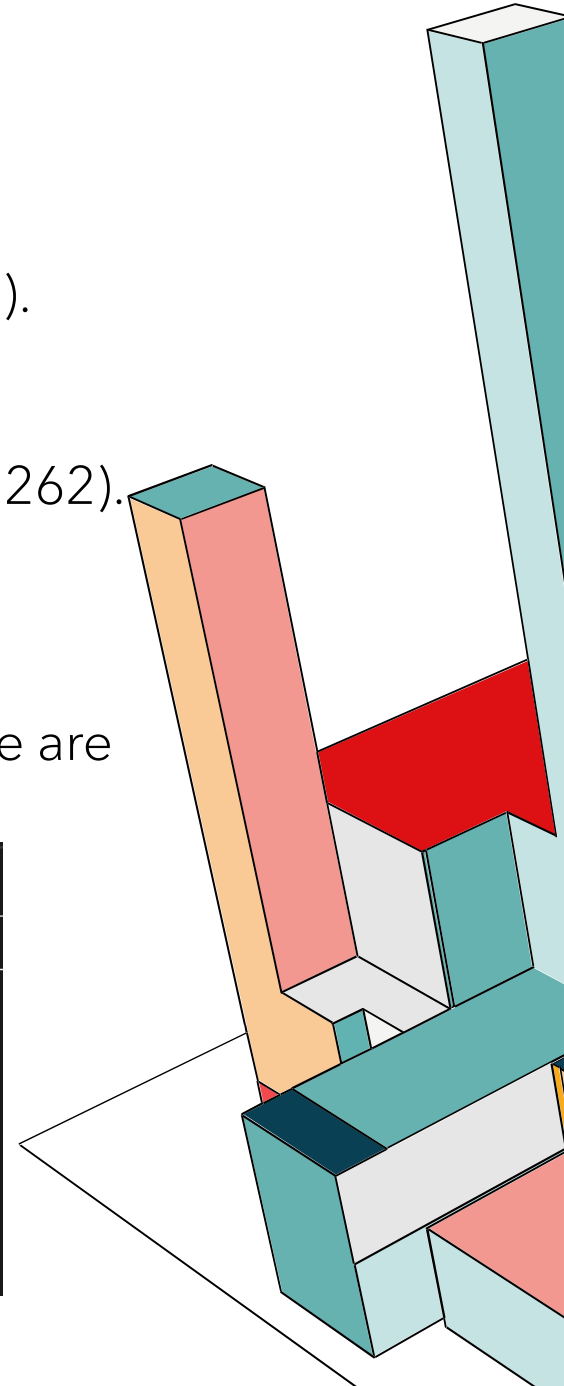
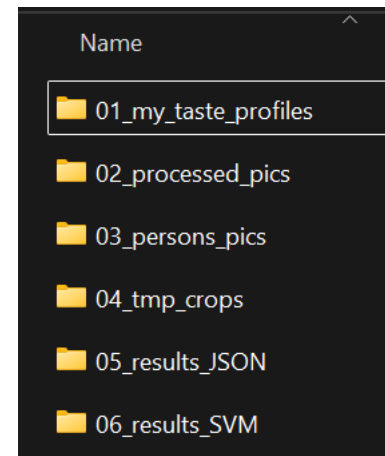
HOW IT WORKS

- **Detector:** YOLO11x (CUDA optimized).
- **Semantic Engine (JSON Pipeline):** Qwen2-VL-7B-Instruct (Multimodal LLM) for descriptive taxonomy mapping.
- **Geometric Engine (SVM Pipeline):** ResNet50 for feature extraction + One-Class SVM (scikit-learn) for boundary classification.
- **Logic:** The JSON pipeline converts pixels into a structured semantic taxonomy (Tier 1 to Tier 4) and scores against a normalized probability matrix, while the SVM pipeline maps image tensors into a multi-dimensional feature space to calculate hyperplane distances.



THE DATA I USED

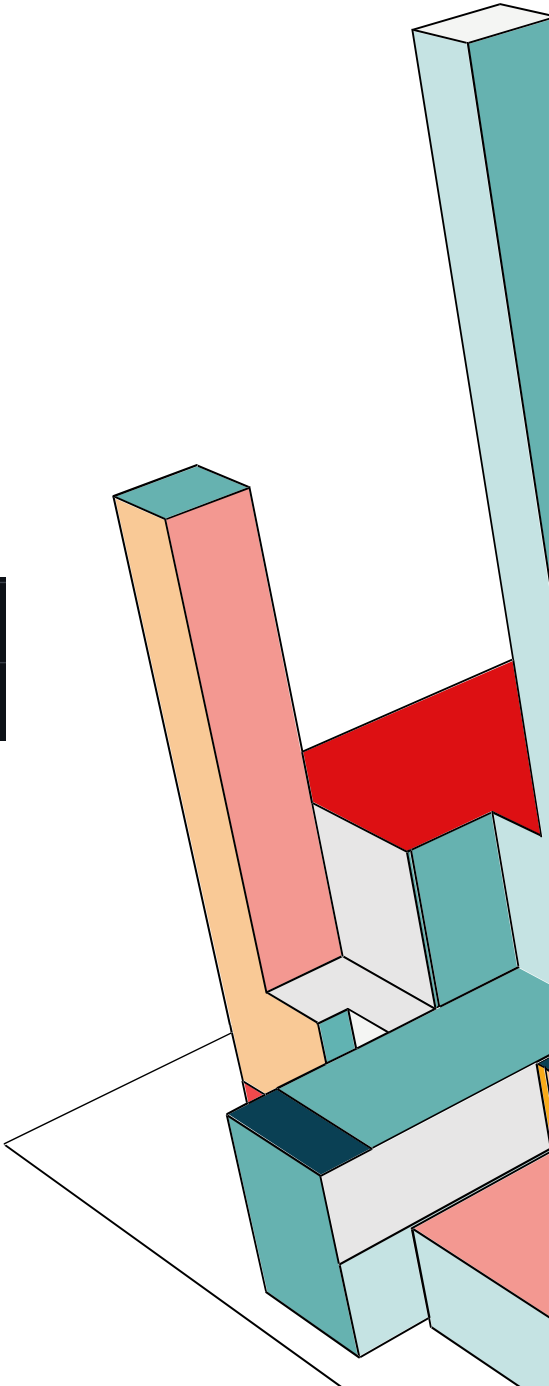
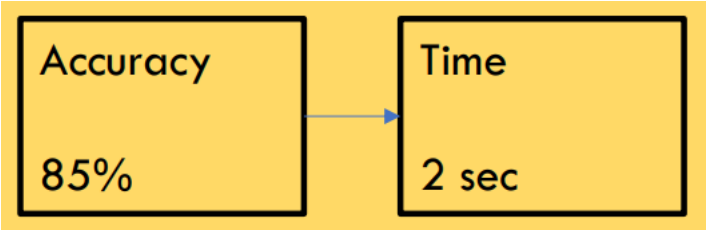
- **Source:** Private photo collection + Synthetic AI generations (Nano Banana).
- **Train** (01_my_taste_profiles): 50 private photos
- **Test** (04_tmp_crops): 10 Target Images, 252 Synthetic Noise Images (Total 262).
- **Labels:** Unsupervised taste clustering (no manual pixel labeling).
- **Cleaning:** Automated filtering using YOLO11x to remove frames without humans and splitting a single image into several images if multiple people are detected



RESULTS VS MY TARGETS

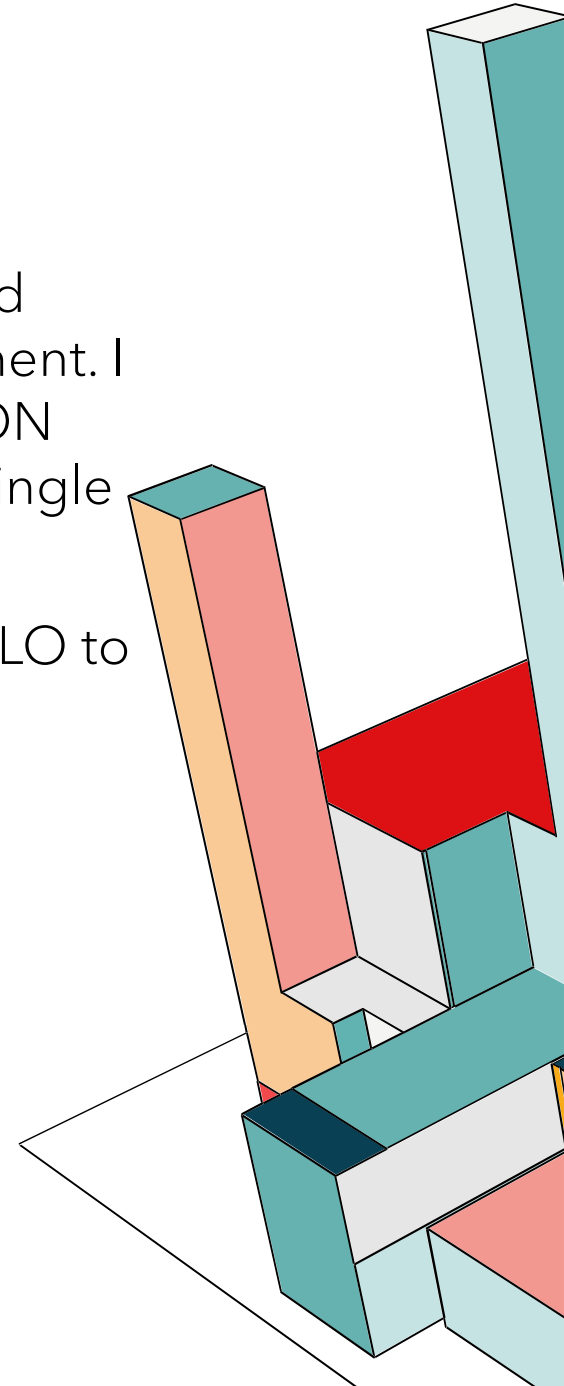
Metric	Blueprint Target	Achieved (JSON Pipeline)	Achieved (SVM Pipeline)
Accuracy	>85.0%	93.89%	92.36%
Recall		70.00%	50.00%
Precision (Optimized)	100.00%	100.00% (Threshold >= 95%)	100.00% (Threshold > 0.05)

System_Performance	Target_Time	< 2.0 sec	Goal for 1 crowd photo	Blueprint requirement for commercial viability
System_Performance	Batch_Scoring_Time	1.8 sec	1.7s (SVM) + 0.1s (JSON)	Execution time to score and rank 262 pre-processed crops



WHAT CHANGED FROM BLUEPRINT

- In that blueprint, I planned to use four different levels for analysis: face and small accessories, body shape, clothes and big accessories, and environment. I changed this plan. I decided to replace those four levels with one big JSON array of features. For that SVM pipeline, I combined all features into one single cloud.
- I planned to test YOLO and maybe SAM 2. I did not use SAM 2. I used YOLO to crop images. I used Qwen2-VL to generate JSON data. I used two other technologies to build that SVM pipeline: ResNet50 and scikit-learn.



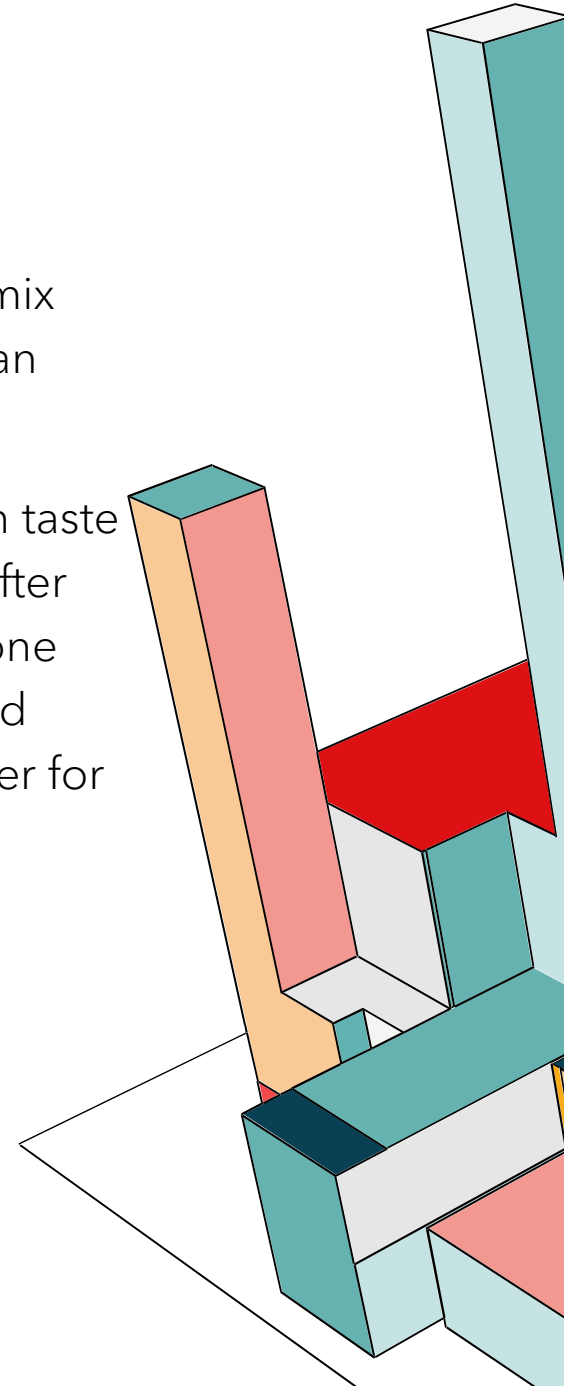
CHALLENGES AND FIXES

- **Problem:** False Positives from AI-generated images breaching the SVM hyperplane.
 - **Fix:** Shifted from pixel-distance calculation to LLM-driven semantic extraction.
- **Problem:** Hardware constraints when processing hundreds of high-res images.
 - **Fix:** Implemented GPU VRAM offloading (`torch.cuda.empty_cache()`) and forced `matplotlib.use('Agg')` backend.



LESSONS AND NEXT STEPS

- **Lessons Learned:** It is very good to combine computer vision with JSON parsing. This mix with SVM pipeline can make accuracy higher. Building this system was much harder than simple lab work. I had many hard bugs. I spent much time to fix them.
- **Next Steps:** I want to make both pipelines more accurate. I will train models not only on taste photos, but also on negative examples. I will add one new folder for wrong matches. After that step, I will build one combined JSON+SVM pipeline. I will connect this system to one web API for real-time scraping and photo sorting. I will also replace heavy YOLO11x and Qwen2-VL-7B-Instruct models with light models. This change will make processing faster for mobile devices. Finally, I want to release one commercial product.



**THANK YOU FOR
THIS CLASS**

